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Simultaneous computation of total viewshed on large high resolution grids

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The knowledge of visibility information on a terrain is essential for a large number of current applications. There exist several algorithms in the literature for building visibility maps (VMs) but only for one single viewpoint or at most for a very small number of observers. This limitation is due to the high computational complexity of the used methods (which is greater than $O(N^2)$, where N is the number of the terrain points) and also due to the fact that single-point algorithms cannot efficiently be scaled to all points. We present a novel fast algorithm able to compute the complete VM, also called total viewshed, where, given a terrain T represented by its regular digital elevation model (DEM), our method simultaneously produces a continuous VM for all the points of T. In particular, the proposed algorithm combines two algorithms, (i) a visible heights algorithm that divides the terrain into sectors and calculates the end of all the visible ring sectors and (ii) an algorithm that finds out the start of those visible ring sectors using a low-cost calculation. Theoretical and experimental comparisons were performed considering the VM of one single observer, because the most used GIS tools compute the viewshed by point. The results demonstrate that our algorithm is faster by many orders of magnitude than the widely used GIS visibility tools showing similar accuracy.

Keywords: digital elevation or terrain models; visibility analysis; geographic information science; spatial analysis; terrain analysis

1. Introduction

In the last decade, new larger digital elevation models (DEMs) of higher precision are being created, for example, a global DEM of the earth's land surface of size 15.1015 pixels and precision 3 arc seconds ($\sim 90 \times 90$ m^2 at the equator) is now available in CGIAR–Consortium for Spatial Information (2011) in addition to a large number of DEMs of many regions of resolution higher than 1 arc second (30×30 m^2). Moreover, in the next 3 years, new global products of higher resolution are expected; TanDEM-X Satellite, launched on 21 June 2010, will allow the generation of global DEMs of resolution higher than 10×10 m^2. These advances are producing a huge need to new fast and efficient geographic information systems (GIS) processing algorithms.

A number of GIS applications on terrains represented by DEMs are concerned with visibility. Examples include the computation of the minimum number of locations that have the largest coverage of a terrain (Franklin and Ray 1994, Ben-Shimo *et al.* 2007) (i) for placing radio, TV, Internet, wireless or mobile phone transmitters and receivers in

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telecommunications, (ii) for situating forest fire watchtowers in environmental planning and (iii) for determining routes for hiking trails in tourism. Recently, visibility analysis is also used for visual impact assessment, such as the impact of aquiculture projects (Perez *et al.* 2003) and wind turbine farms installations (Kidner *et al.* 1997). It is also used to improve size estimation techniques wildlife population of (Maichak and Schuler 2004), and in the field of archeology, it is used for the reconstruction of views to and from historical objects and sites (Ogburn 2006).

According to Llobera (2003), viewshed algorithms can be divided into two categories depending on the target application. The first focuses on studying urban and building spaces, where as the second analyzes natural landscapes. Most recent approaches in the first category are based on the concept of isovist and isovist fields introduced by Benedikt (1979). Our algorithm belongs to the second category.

In this context, most visibility software, such as the well-known Viewshed tool implemented under the commercial ArcGIS 10 (2010) Environmental systems Research Institute and r.los implemented under GRASS GIS 6.4 (GRASS 2011), and several algorithms provided in the literature (Atallah 1983, Cabral *et al.* 1987, De Floriani *et al.* 1989, Hershberger 1989, De Floriani and Magillo 1994, Miller *et al.* 1995, Miller 2001, De Floriani and Magillo 1993, Katz *et al.* 1991) compute the parts of a terrain that are visible from a single viewpoint or at most from a reduced number of observers. However, none of these implementations is capable of computing the total viewshed simultaneously, that is, viewshed at all the points of the terrain in one step. The reason behind this limitation is that the calculation of visibility map (VM) even for a single point is very expensive and has a complexity greater than $O(N)$, where N is the total number of points of the DEM. This means that computing the complete VM at all the points of the terrain has a complexity greater than $O(N^2)$. Some works propose solving this limitation by parallel computing of simple single point-based viewshed algorithms on multiprocessor systems (Llobera *et al.* 2010, Mineter *et al.* 2003).

This article presents a novel algorithm able to compute the complete VM of the entire terrain at a low computational cost. In contrast to most previous algorithms that process all the N points of the DEM in order to produce the VM of one single point, our method finds out the visible points by regions. That is, it looks for the visible points only among a selected set of points, called sector. As the size of the selected set is $O(\log(N/S))$, finding the visible regions over all the terrain has a complexity of only $O(S.N.\log(N))$, where S is the total number of sectors. As far as we know, this algorithm is the first in simultaneously computing the VM for all the points of the terrain. To make our algorithm comparable with the most used visibility analysis tools, that is, r.los (under GRASS) and Viewshed tool (under ArcGIS), we compared the accuracy and run-time of computing VM for individual viewpoints. The experimental results demonstrate that our algorithm is significantly faster than r.los (under GRASS) and Viewshed tool (under ArcGIS) showing similar accuracy.

This article is organized as follows: A brief description of related works is given in Section 2. A detailed description of all the algorithmic strategies, necessary for carrying out both the stages of the algorithm, is provided in Section 3. Numerical and computational results are analyzed and discussed in Section 4, and finally, conclusions are given in Section 5.

2. Related works

The problem of computing continuous VM with respect to a single viewpoint has been extensively studied in the literature. Nagy (1994) and Floriani and Magillo (2003) provide

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complete surveys of visibility algorithms. In this section, we will discuss only those papers more related to our own work.

Atallah (1983) uses a divide-and-conquer algorithm to compute the upper envelope of N segments in the plane with a time complexity of $O(N\alpha(N)\log N))$, where α is the inverse Ackermann function, a slow growing function, nearly constant. This algorithm recursively divides the line segments into halves and pairwise merges the resultant upper envelopes via a sweep line technique. Hershberger (1989) improves Atallah's algorithm, reaching an optimal complexity of $O(N\log N)$. De Floriani *et al.* (1989) computed the VM on triangulated irregular networks (TINs) using the front-to-back order in time $O(N^2\alpha(N))$.

Alternatively, the visibility problem can also be solved using horizon calculation. The approximative method of Cabral *et al.* (1987) divides the horizon into S sectors, and in each sector, it determines the elevation of the horizon, considering solely the points of the terrain that are in the central line of the sector with a computational cost $O(S(N^{1.5}))$. Stewart (1998) proposed a more precise and faster algorithm that computes the approximate horizon for all the points of the DEM in the whole sector with a total cost, $O(SN(\log^2 N + S))$, and space, $O(N(\log N))$. Finally, Tabik *et al.* (2011) proposed a three-level horizon approach more than three times faster than Stewart's algorithm while maintaining the same accuracy. Nevertheless, none of these works produces the VM at all the points of a terrain at the same time.

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3. The proposed algorithm

Our algorithm works in two main phases. Actually, it solves the visibility problem by finding the limits where the visible regions start and end. In the first phase, it searches for the end limits of the visible regions, of ring sector shapes, in each direction around the points of the terrain. In the next phase, it uses a fast algorithm to calculate the start limits of those visible ring sectors.

In this section, we provide the notion of visible ring sectors, the algorithmic basis of the first stage of the algorithm, the optimizations that make the whole algorithm faster and finally a summary.

3.1. Notion of visible ring sector

We formulate VM computation to determine where the continuous visible regions start and end. We first partition the space around the points of the terrain into sectors and then proceed to compute the visible areas, of ring sector shapes, in each sector s. A visible ring sector from a given point P is delimited by the set of points where the ring sector starts, which we call Start of Ring Sector SRS.s.P, and the set of points where the ring sector ends, which we call End of visible Ring Sector ERS.s.P (See Figure 1 for illustration). The sector should be very narrow so that the projection of the points of the sector onto the vertical plane that passes from z-axis can be considered as a single profile. Very narrow sectors ensure a high accuracy of the method.

3.2. Basis of the profile computation: hull tree

The DEM of a terrain T can be described as a set of N points P_i ($i = 0, \ldots, N$) of coordinates (x_i, y_i, z_i), where z_i is the height of the point. The algorithm divides the space around the points P_i of the DEM into S sectors of azimuth angle $\frac{2\pi}{S}$ radians each one. Then, it

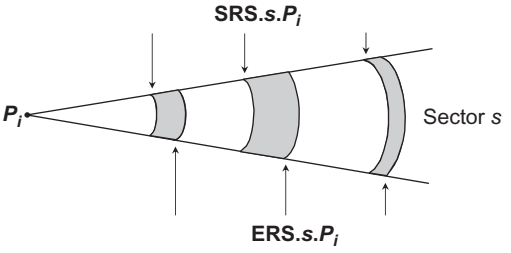


Figure 1. Point P_i has three visible ring sectors (in light gray) in the region delimited by sector s. These visible ring sectors are determined by the set of Start of Ring Sector points SRS.$s.P_i$ and the End of Ring Sector points ERS.$s.P_i$.

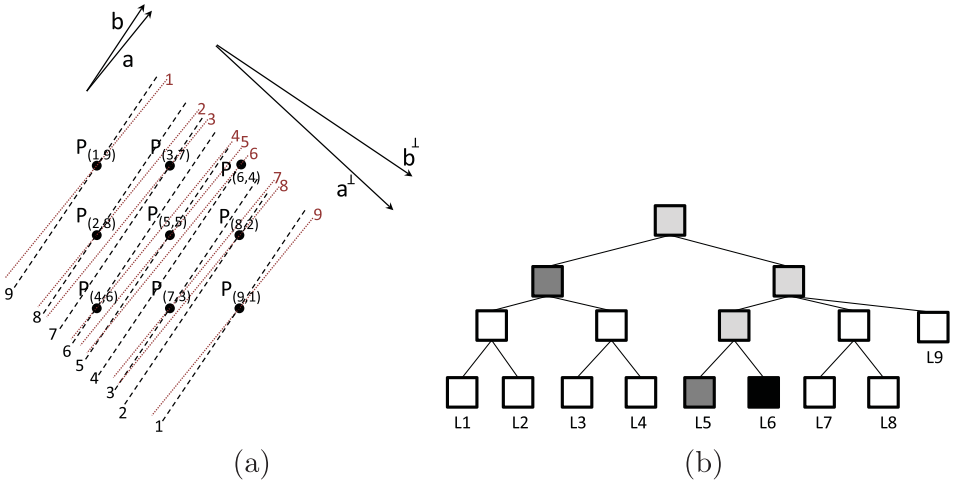


Figure 2. (a) For each sector s delimited by vectors **a** and **b**, the points of the DEM are sorted by $\mathbf{a}^{\perp}$ and $\mathbf{b}^{\perp}$. (b) The hull tree contains as many leaves as the number of points in the DEM, each leaf represents a point and each node is a hull that covers its own leaves (i.e., sons). To process point L6, in black color, the search of the ERS points is made among the points in light gray color and simultaneously L6 is added as possible ERS to nodes in dark gray color.

iterates over all the S sectors. At the end of this step, it obtains all the ERS points of all visible regions for all points.

In particular, for each s iteration, the algorithm reorders the points in a new coordinate system, $(\mathbf{a}^{\perp}, \mathbf{b}^{\perp}, z)$, where $\mathbf{a}^{\perp}$ and $\mathbf{b}^{\perp}$ are two vectors perpendicular to vectors **a** and **b** that delimit sector s, as can seen in Figure 2a. The algorithm processes the points in $\mathbf{a}^{\perp}$ order in such a way that each point finds its profile in a complex data structure, called Hull Tree (HT), and simultaneously it is incorporated as a candidate profile point for the points that will be processed in next iterations. More details about the HT can be found in Stewart (1998) and Tabik (2011).

HT is a binary tree of N leaves. Each leaf represents a point and each node is a convex hull that covers its own leaves (i.e., sons), which is shown in Figure 2b. The structure and processing order of the tree ensure a highly efficient computation as the profile key information is always found among the $O(\log N)$ leaf-to-root nodes in the tree. It will be shown in Section 3.4 that HT contains all the key points necessary to accurately compute the limits of the visible regions.

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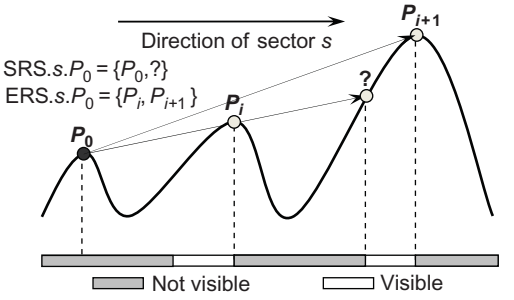


Figure 3. In direction s, P_i and P_{i+1} belong to the set of end-of-visible-ring sectors points of P_0 (i.e., ERS.$s._{P_0} = \{P_i, P_{i+1}\}$). While in the opposite sector, s', with $\hat{s}' = -\hat{s}$, P_i and P_0 belong to the End-of-visible-Ring sectors of the point labeled as "?" (i.e., ERS.s'. ? $= \{P_i, P_0\}$. The point labeled as "?" belongs to the set of start-of-visible-ring sectors of P_0 and it is found in the second stage of the algorithm.

3.3. *Opposite sector-based optimization*

In each sector s and for each point P of the terrain, the sets SRS.s.P and ERS.s.P must be found. Suppose the algorithm has already calculated the ERS.s.P of a specific point P in sector s. In future iterations, particularly in the opposite sector s' such as $\hat{s}' = -\hat{s}$ (i.e., $\hat{s}' = \hat{s} + \pi$), the algorithm will analyze again the point P as possible SRS.s', in sector s', of its own ERS.s.P points. For simplicity, consider the easily drawn example in Figure 3. If the point labeled as "?" belongs to SRS.s.P_0, in direction s, thus P_0 may belong to ERS.s'.? in the opposite sector s'. Therefore, we first compute and store the ERS of all the points of the terrain and then in a second stage use a fast postprocessing to obtain the SRS points. This approach reduces the storage requirements and calculation by half.

3.4. *End-of-ring sectors*

Our algorithm searches for the ERS.s.P_i among the profile points based on the following assumption. The ERS points of P belong to the set of points that form P's own convex profile sets (or hulls) in s. This can be demonstrated mathematically as follows:

Lemma: $\forall Q \in HT$, *if* $Q \in ERS.s.P \Rightarrow Q \in$ *some hull*.

Proof: suppose $Q \in$ ERS.s.P and $Q \notin$ any hull.
this means that $\exists! Q'$ such as Q' elevation is larger than P elevation
which implies that $Q \notin$ ERS.s.P

To determine the ERS.s.P points among the profile points, the algorithm processes the profile points in the increasing distance order from P. If the tangent between the projection of the last profile point and the current profile point is greater than the maximum angle, the current point is added to the ERS.s.P set.

3.4.1. *ERS computation algorithm summary*

In summary, the first stage of the algorithm calculates the ERS as follows:

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For each sector s in S

- We sort all the points of the DEM by $a^{\perp}$ and $b^{\perp}$ in the new system $(a^{\perp}, b^{\perp}, z)$. Since the target DEMs are regular grids, the sorting operation can be carried out by counting the number of points behind the sweep line using very simple trigonometric operations, i.e., the position number is directly computed. In practice the time spent in this phase is negligible. We reuse the ordering by $b^{\perp}$ in sector i to reorder by $a^{\perp}$ in sector $i + 1$. The new indices j and k are determined for each point.
- For each point $P_{j,k}$ in order of increasing index j:
  - Compute the profile points using the Hull tree of $P_{j,k}$.
  - Add to tree new candidate as possible profile for next points. The tree is updated by inserting the heights of the new point $P_{j,k}$. Index k is used to insert the point in the structure.
  - Find ERS points among profile points and store them in the ERS structure.
- The points are reordered from their indices j to recover the original ordering.

3.5. Start-of-ring sector calculation

Once all the ERS points are calculated for all the points of the terrain, the algorithm proceeds to compute the SRS points by sector and by point. In this stage, each point P searches for the points that are at the end of ring sector and stores them into a data structure that we call reverse list. In principle, all the points that belong to this list are candidates for SRS of P. For illustration consider point P_0 in Figure 3. Among the set of points limited by two consecutive ERS points, P_i and P_{i+1}, the points that have P_0 as ERS in the opposite sector and belong to the interval $[P_i, P_{i+1}]$ are candidates for SSR.s.P_0. Only the point that verifies the following criteria is SRS of P_0. It is farther than P_i and nearer than P_{i+1}, and obviously it is a candidate in the reverse list.

This stage is very slight from a computational point of view. For instance, for the terrain shown in Figure 4(a) of size 2000×2000 points, the second stage takes only 33 minutes on a Desktop based on Intel core2 duo that runs at 3.16 GHz with 3.25 GB of main memory. Actually, computation of reverse list takes 3 minutes and calculation of SRS takes 30 minutes.

4. Numerical and computational results

Some numerical results of our approach are shown in Figure 4. In particular, Figure 4b shows the VM (or total viewshed) computed on the terrain of the city of Malaga, Spain, whose DEM is shown in Figure 4a. The value at each pixel in Figure 4b shows the number of visible km^2. Dark red color corresponds to points with very high visibility, about 100 visible km^2. Dark blue color corresponds to points with very low visibility. From this figure, one can observe that higher points do not necessarily have higher visibility. Actually, some points at the sea level have notably more visibility than higher interior points. The advantage of our method over other algorithms is that it provides a global comparison of the visibility over all the points of the terrain.

The precision of the VM calculated by our algorithm depends on the resolution of the used DEM and the considered number of sectors. It can be increased or decreased depending on the needs of the considered application. Notice that the azimuthal sectorization of

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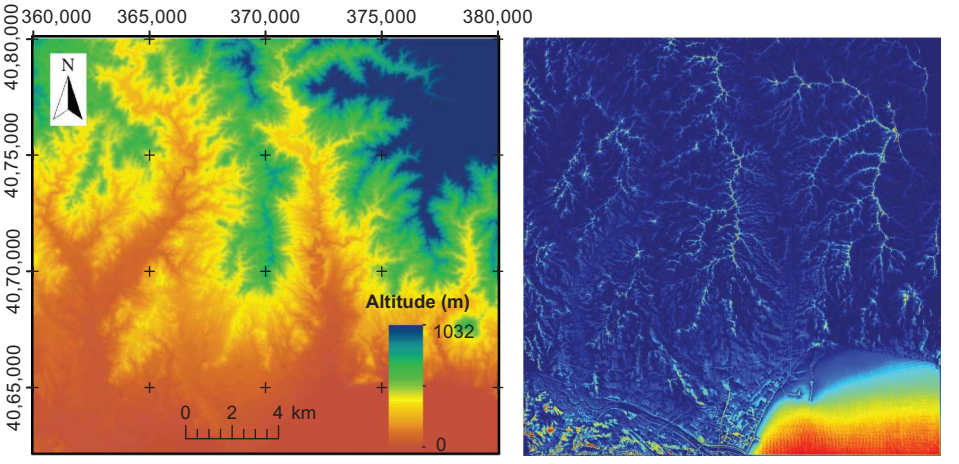


Figure 4. (a) Elevation map of the city of Malaga, Spain, of size 2000 × 2000 points and resolution $10 \times 10\ m^2$. (b) The total viewshed of the city of Malaga; dark red and dark blue colors show higher and lower numbers of visible km^2, respectively.

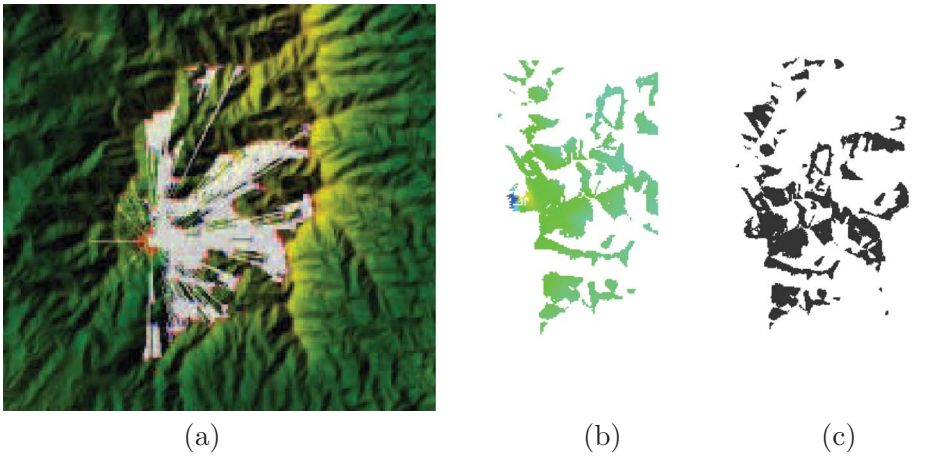


Figure 5. The viewshed calculated in the point of UTM (zone 30S) coordinates, $X = 368.800$ and $Y = 4.070.140$, in a DEM of resolution $10 \times 10\ m^2$, using (a) our model (the light gray regions indicate the visible ring sectors), (b) r.los under GRASS (the green color indicates visible points), and (c) Viewshed tool under ArcGIS (the black color indicates visible points).

the viewshed involves a slight linear loss of precision in very distant locations. This fact can be appreciated in Figure 5, which shows the visible ring sectors seen at a point selected from the area of the DEM of the city of Malaga. However, these kinds of precision losses are irrelevant in most applications, because signals, intensity is inversely proportional to the square of the distance ($I = P/4\pi d^2$, where P is the power of the signal). Additionally, one can appreciate the similarity of the viewsheds calculated by our method, r.los and Viewshed tool shown in Figure 5(a–c), respectively.

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4.1. Comparison between r.los, Viewshed tool, and our method

Several studies that compared actual and predicted viewsheds demonstrated that the widely used viewshed methods have a low average accuracy that hardly surpased 50% (Ruiz 1995, Maloy and Denis 2001). These studies evidenced that the accuracy of viewshed algorithms depends mainly on the accuracy of the DEM, that is, the higher the resolution of the DEM the higher the accuracy of the calculated viewshed. In addition, this problem is unstable numerically, because insignificant variations in the observer location (i.e., in its latitude, longitude, altitude, etc.) may generate very large variations in its viewshed. All these factors make comparisons between viewshed methods a very hard task.

To compare the accuracy of our model with respect to r.los (under GRASS) and Viewshed tool (under ArcGIS), we calculated the number of visible km^2 from 32 sample points. We only show and discuss the results of 29 sample points because r.los was not able to produce the viewshed of three of them.

Figure 6 plots the relative differences of each one of the three tools with respect to the mean value calculated as:

$$\text{Relative difference}(\text{model}_i) = \frac{\#\text{visible km}^2(\text{model}_i) - \text{meanValue}(\text{viewshed, r.los, ours})}{\text{meanValue}(\text{viewshed, r.los, ours})}.$$

As can be observed in Figure 6, the three models provide similar values for about 65.5% of the total number of points with a relative difference $\in [-20\%, +20\%]$, which can be considered reasonable due to the instability of the problem. Besides, our algorithm gives the nearest value to the mean value for point 1. However, r.los and Viewshed tool obtain similar values for points 19, 24, 25, and 29, whereas our model shows a number of visible km^2 significantly higher. This is due to the effect of narrow sectors intrinsic to our method (i.e., it uses a very small sector angle $= 1°$). That is, for very instable viewpoints it may occur that the nearest elevations outside the considered sector may be neglected (Ruiz 1995). Due to the instability of the problem, the calculated viewsheds of these instable points, using the three tools, correspond in fact to viewsheds of real points in the closer surrounding area. In consequence, this is translated into a slight shift, of pixel scale, of these viewpoints in the final total viewshed map.

Our algorithm can optionally solve the narrow sector effect by performing an interpolation based on a set of the nearest surrounding points although it does not produce

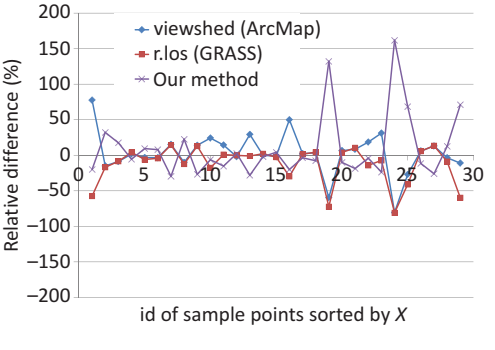


Figure 6. Relative difference, in %, between Viewshed tool (ArcGIS), r.los (GRASS), and our model with respect to the mean value of the three models.

visible differences, which can be seen in Figure 5(a–c). The cost of this additional calculus increases the run-time complexity by about 20%.

4.2. Run-time comparisons

The code was implemented in C++ and compiled with -O3. For performance evaluations, we compared the run-time of our algorithm versus the run-time of r.los (under GRASS) and Viewshed tool (under ArcGIS), a widely used software for viewshed calculation. We performed all experiments on a PC based on Intel core2 duo running at 3.16 GHz with 3.25 GB of main memory. Computing the ERS points of all the points of a 2000 × 2000 points DEM (of size 8 MB), considering 360 sectors, takes about 6.5 hours. Building the reverse list takes 3 minutes and finding the SRS points takes about 30 minutes. Storing the ERS points by sector in files takes 10 minutes resulting in a set of 360 files of a total size of about 42 GB. This means that the storage requirements per point per sector is about 30 bytes. In the VM of the city of Malaga shown in Figure 4b, the average number of ring sectors per point per sector is equal to 3.1 ring sectors.

Taking into account all these measurements, computing the VM at a single point on a terrain of size 2000 × 2000 points using the algorithm we proposed in this work takes only 0.0063 seconds, while Viewshed tool and r.los take about 10 and 155 seconds, respectively, to compute VM in the same conditions.

5. Conclusions

We presented a new and fast algorithm able to compute the total viewshed simultaneously for all the points of the terrain. The proposed algorithm can be very useful for accelerating a wide range of visibility-based applications. For example, it can be used to accelerate sitting multiple observers so as to jointly cover as much terrain as possible described in Franklin (2002) and for extending the visibility index calculation, cited in Franklin *et al.* (1994), (2002), to a very large radius. The presented algorithm has a time complexity $O(S.N.\log(N))$, substantially better than previous single-point-visibility algorithms and scales $O(N.\log(N))$ versus $O(N^2)$ for previous algorithms. Moreover, experimental results demonstrate that our method is faster by many order of magnitude than mostly used visibility tools including GRASS and ArcGIS showing similar accuracy.

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